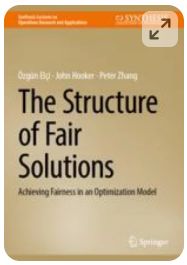
SPRINGER NATURE Link

Log in



The Structure of Fair Solutions

Achieving Fairness in an Optimization Model

| Book | © 2026

Accessibility Information

Overview

Authors: Özgün Elçi , John Hooker , Peter Zhang

Presents practical linear, nonlinear, or mixed integer programming formulations and a wide variety of fairness models

Includes detailed proofs that provide insight into the properties of each criterion

Provides guidelines for selecting a fairness model and the tendency to incentivize cooperation or competition

Part of the book series: Synthesis Lectures on Operations Research and Applications (SLORA)

1340 Accesses 4 Citations 7 Altmetric

This is a preview of subscription content, log in via an institution to check access.

Access this book

Log in via an institution

eBook USD 34.99
Price excludes VAT (USA)

Available as EPUB and PDF
Read on any device
Instant download
Own it forever

Buy eBook

Softcover Book USD 44.99

Hardcover Book USD 44.99

Tax calculation will be finalised at checkout

Other ways to access

Licence this eBook for your library →

Institutional subscriptions →

About this book

This book provides a novel and unifying perspective on the structural properties of fair solutions in optimization formulations. The book also addresses a growing interest in incorporating fairness into the models that lie behind many business and public policy decisions. Since there are several ways to formulate fairness mathematically, the authors characterize optimal solutions that result from different formulations with the aim of informing the choice of an appropriate model for a given application. The focus is on fairness criteria that combine efficiency with fairness since typically both are important in practice. Most of these results are new and do not appear in the current literature. The book is directed towards a wide range of audiences including practitioners, researchers in mathematical optimization, and welfare economists.

In addition, this book:

- Presents practical linear, nonlinear, or mixed integer programming formulations and a wide variety of fairness models
- Includes detailed proofs that provide insight into the properties of each criterion
- Provides guidelines for selecting a fairness model and the tendency to incentivize cooperation or competition

About the Authors

Özgün Elçi, Ph.D., is a Research Scientist on the Modeling and Optimization team at Amazon.

John Hooker, Ph.D. is University Professor of Operations Research and T. Jerome Holleran Professor of Business Ethics and Social Responsibility, Emeritus, at Carnegie Mellon University.

Peter Zhang, Ph.D, is an Assistant Professor at Carnegie Mellon University's Heinz College of Information Systems and Public Policy.

Explore related subjects

Discover the latest articles, books and news in related subjects.

Analysis Business Ethics Optimization

Operations Research and Decision Theory Stochastic Networks

Operations Research, Management Science

Inequality Measurement and Statistical Analysis

Search within this book

Search

Table of contents (15 chapters)

Front Matter

Pages i–ix

Download chapter PDF

Introduction

Özgün Elçi, John Hooker, Peter Zhang

Pages 1–7

Optimizing with Fairness

Özgün Elçi, John Hooker, Peter Zhang
Pages 9–15

Hierarchical Distribution

Özgün Elçi, John Hooker, Peter Zhang
Pages 17–19

Incentives and Sharing

Özgün Elçi, John Hooker, Peter Zhang
Pages 21–22

Utilitarian and Maximin Criteria

Özgün Elçi, John Hooker, Peter Zhang
Pages 23–26

The Leximax Criterion

Özgün Elçi, John Hooker, Peter Zhang
Pages 27–31

Beta Fairness

Özgün Elçi, John Hooker, Peter Zhang
Pages 33–37

Alpha Fairness

Özgün Elçi, John Hooker, Peter Zhang
Pages 39–45

Kalai–Smorodinsky Bargaining

Özgün Elçi, John Hooker, Peter Zhang
Pages 47–51

Utility Threshold Criterion

Özgün Elçi, John Hooker, Peter Zhang
Pages 53–61

Equity Threshold Criterion

Özgün Elçi, John Hooker, Peter Zhang
Pages 63–71

Utility Threshold Criterion with Leximax Fairness

Özgün Elçi, John Hooker, Peter Zhang
Pages 73–82

Inequality Bounds: Range

Özgün Elçi, John Hooker, Peter Zhang
Pages 83–86

Inequality Bounds: Gini Coefficient

Özgün Elçi, John Hooker, Peter Zhang
Pages 87–94

Conclusions

Özgün Elçi, John Hooker, Peter Zhang
Pages 95–104

Back Matter

Pages 105–108

Download chapter PDF

Back to top

Authors and Affiliations

Amazon, Bellevue, USA
Özgün Elçi

Tepper School of Business, Carnegie Mellon University, Pittsburgh, USA
John Hooker

Carnegie Mellon University, Pittsburgh, USA
Peter Zhang

About the authors

Özgün Elçi, Ph.D., is a Research Scientist on the Modeling and Optimization team at Amazon. He has a Ph.D. in Operations Research from Tepper School of Business at Carnegie Mellon University. During his doctoral studies, he worked on applying logic-based Benders decomposition to stochastic scheduling problems. He also worked on robust portfolio optimization and fair allocation of resources. Prior to that, he focused on humanitarian relief network design problems under

uncertainty. His current research interests include solving large-scale network design problems that arise in industry via decomposition-based exact methods.

John Hooker, Ph.D. is University Professor of Operations Research and T. Jerome Holleran Professor of Business Ethics and Social Responsibility, Emeritus, at Carnegie Mellon University. He holds doctoral degrees in philosophy and operations research. He has published 200+ articles, 10 books, and six edited volumes in operations research, constraint programming, ethics, the mathematical analysis of fairness, cross-cultural management, and music theory. He is a Fellow of the Institute for Operations Research and the Management Sciences (INFORMS), as well as recipient the INFORMS Computing Society Prize, the INFORMS Khachiyan Award for lifetime achievements in optimization, the Research Excellence Award of the Association for Constraint Programming, and several best paper awards. He is a pioneer in the integration of optimization and constraint programming technologies, now a feature of leading optimization solvers. He developed a substantial generalization of Benders decomposition that has seen hundreds of practical applications. In addition, he and Tarik Hadžić introduced an approach to optimization based on decision diagrams. Presently, his main research area is the mathematical analysis of fairness. His ethics-related books include *Business Ethics as Rational Choice*, *Working across Cultures*, *Taking Ethics Seriously*, *Advanced Introduction to Business Ethics, and Toward Systematic Normative Ethics (forthcoming)*. He served as founding editor-in-chief of the *Journal of Business Ethics Education*, the only academic journal dedicated to the topic.

Peter Zhang, Ph.D, is an Assistant Professor at Carnegie Mellon University's Heinz College of Information Systems and Public Policy. His research interests include robust and stochastic optimization theory, with applications in supply chain and transportation. He obtained Ph.D. from MIT in Engineering Systems and BASc and MASc from the University of Toronto. His Ph.D. thesis contributed to the

identification of vulnerabilities in supply chain networks and robust supply chain network design methods. Some concepts that Peter worked on and developed at MIT (time-to-recover, supply chain stress test, and robust supply chains) have been cited in various media outlets, academic journals, and White House reports. He has received multiple paper competition awards in academic and applied research areas at INFORMS and POM conferences.

Accessibility Information

PDF accessibility summary

This PDF has been created in accordance with the PDF/UA-1 standard to enhance accessibility, including screen reader support, described non-text content (images, graphs), bookmarks for easy navigation, keyboard-friendly links and forms and searchable, selectable text. We recognize the importance of accessibility, and we welcome queries about accessibility for any of our products. If you have a question or an access need, please get in touch with us at accessibilitysupport@springernature.com. Please note that a more accessible version of this eBook is available as ePub.

EPUB accessibility summary

This ebook is designed with accessibility in mind, aiming to meet the ePub Accessibility 1.0 AA and WCAG 2.2 Level AA standards. It features a navigable table of contents, structured headings, and alternative text for images, ensuring smooth, intuitive navigation and comprehension. The text is reflowable and resizable, with sufficient contrast. We recognize the importance of accessibility, and we welcome queries about accessibility for any of our products. If you have a question or an access need, please get in touch with us at accessibilitysupport@springernature.com.

Bibliographic Information

Book Title
The Structure of Fair Solutions

Book Subtitle
Achieving Fairness in an Optimization Model

Authors
Özgün Elçi, John Hooker, Peter Zhang

Series Title
Synthesis Lectures on Operations Research and Applications

DOI
https://doi.org/10.1007/978-3-031-82190-5

Publisher
Springer Cham

eBook Packages
Synthesis Collection of Technology (R0)

Copyright Information
The Editor(s) (if applicable) and The Author(s), under exclusive license to Springer Nature Switzerland AG 2026

Hardcover ISBN
978-3-031-82189-9
Published: 11 July 2025

Softcover ISBN
978-3-031-82192-9
Published: 12 July 2026

eBook ISBN
978-3-031-82190-5
Published: 10 July 2025

Series ISSN
2770-6303

Series E-ISSN
2770-6311

Edition Number
1

Number of Pages
IX, 108

Number of Illustrations
10 b/w illustrations, 11 illustrations in colour

Topics
Operations Research, Management Science, Operations Research/Decision Theory, Optimization, Analysis, Business Ethics, Probability Theory and Stochastic Processes

Keywords

Fairness in Optimization
Fairness and Efficiency
Welfare Optimization
Hierarchical Distribution
Alpha Fairness
Maximin Criteria
Leximax Criteria
Nash Bargaining Solution

Publish with us

Policies and ethics

Back to top